STAT2004 – Tutorial Week 9

We will cover Exercises 1 and 3 together in the tutorial. Please complete Exercises 2 and 4 for homework, and check your solutions with your tutor next week.

Exercise 1 (Robustness of t-methods)

t-based methods are derived under the assumption of normal data. However, the main reason why t-methods have proven so useful in almost all areas of applied science is that they are remarkably robust to departures from normality, even with moderate sample sizes. In this exercise, we will test the practical performance of the classical t-based confidence interval when applied to Poisson data, which are quite evidently *not* normal (why not?).

First, open up `R studio` and set the working directory via `Session > Set Working Directory > Choose Directory...` Choose a directory that you have access to (e.g., your USB disk or UQ network drive).

(a) **Work in pairs for parts (a)-(b)** . Using the following `R` code as a template, generate a random sample of Poisson data using a **population mean $\mu > 0$ of your choice** and a small/moderate **sample size n of your choice**.
Do NOT let your partner know what your chosen population mean is.

```
#########################################################################
n = 22          # choose your own sample size between 15 and 30
mu = 8.7654     # choose your own popn mean > 0

# generate a random sample from Poisson distribution
mydata = rpois(n, mu)

# save your dataset as a .csv (comma separated value) file
write.table(mydata, "Alan.csv", sep=",", col.names= "data", row.names=F)
#########################################################################
```

Email your .csv dataset to your partner.

(b) Download your partner's dataset to your working directory. Using the following `R` code template, load your partner's dataset, calculate the sample mean, sample standard deviation and compute a classical t-based 90% confidence interval for the true population mean.

```
#########################################################################
# load your partner's dataset into R
Alan = read.csv("Alan.csv", header = T)
```

```
# quickly inspect your partner's dataset; does it look like normal data?
Alan$data

# compute sample mean and sample standard deviation
# of your partner's dataset
xbar = mean(Alan$data)
s = sd(Alan$data)

# compute the lower and upper bounds for
# a 90% t-based confidence interval for the popn mean
margin.of.error = qt(0.95, df=n-1)*s/sqrt(n)
lower = xbar - margin.of.error
upper = xbar + margin.of.error
print(c(lower, upper))
##########################################################################
```

Turn to your partner and tell them: "I am 90% confident that the true population mean that you chose is between ... and ..."

Do NOT let your partner know whether their confidence interval was successful in containing your true population mean. This is because we would never know what the true value is in practice. However, you can tell your tutors discreetly (either in-person or via email).

(c) Of course, we can't make any general claims about the robustness of the t-based confidence intervals based on only one or two samples. We need to repeat this confidence interval procedure over many, many samples to see whether these so-called "90% confidence intervals" actually work 90% of the time! Use the following `R` code as a template to examine this.

```
##########################################################################
# repeat parts (a) and (b) across 1,000 different random samples to see
# if t-based 90% CIs actually work 90% of the time with Poisson data

N = 1000    # number of repeated samples

# empty vector to store whether the 90% CI covered the true mean or not
coverage90 = vector()

n = 15         # choose your own sample size between 15 and 30
mu = 16.789    # choose your own popn mean

for(i in 1:N){
  # generate a sample from Poisson distribution
  mydata = rpois(n, mu)

  # compute sample mean and sample standard deviation
```

```
  xbar = mean(mydata)
  s = sd(mydata)

  # compute the lower and upper bounds for
  # a 90% t-based confidence interval for mu
  margin.of.error = qt(0.95, df=n-1)*s/sqrt(n)
  lower = xbar - margin.of.error
  upper = xbar + margin.of.error

  # check whether 90%CI from this particular sample contains true mean
  coverage90[i] = (mu >= lower)*(mu <= upper)
}

# Did the 90% t-based confidence intervals cover the true mean
# in (approx) 90% of samples, even though data were Poisson?
mean(coverage90)
###########################################################################
```

Try different values of the population mean and/or different sample sizes. Does the 90% $t-$based confidence interval still have the correct coverage in all these different scenarios?

Exercise 2 (Significance level, power, type I and type II errors)

Are each of the following sentences true or false? If a sentence is false, modify it appropriately so that it becomes true. [This question is adapted from Question 9.12.5 from Rice (1995) "Mathematical statistics and data analysis"]

(a) The significance level of a test is equal to the probability that the null hypothesis is true.
(b) If the significance level of a test is decreased, the power would be expected to increase.
(c) If a test is rejected at significance level α, then the probability that the null hypothesis is true equals α.
(d) The probability that the null hypothesis is falsely rejected is equal to the power of the test.
(e) A type I error occurs when the statistic falls in the rejection region of the test.
(f) A type II error is more serious than a type I error.
(g) The power of a test is determined by the null distribution of the test statistic.
(h) The p-value is a random variable.

Exercise 3 (Testing for a biased coin)

A coin is to be thrown independently 10 times to the test the hypothesis that the coin is fair versus the alternative that the coin is biased.

(a) Identify the parameter of interest in this test, and write down the null and alternative hypotheses in terms of this parameter.

A decision rule is proposed to reject the null hypothesis if either 0 or 1, or 9 or 10 heads are observed.

(b) What is the significance level of this test?
(c) If in fact the probability of heads is 0.1 (i.e., the coin is severely biased), what is the power of this test?
(d) Compute and plot the power curve of this test across the entire parameter space.

Exercise 4 (Testing for a loaded die)

We suspect a certain die to be loaded. Rolling it 100 times we observe 25 sixes. Is there enough evidence to justify our suspicion?

(a) Identify the parameter of interest here, and write down the null and alternative hypotheses in terms of this parameter.
(b) Compute the p-value of this test, and make a decision between the two hypotheses you specified in part (a).

Exercises involving biased coins and loaded dice are often dismissed as merely esoteric musings for probability purists, with little application to the *real* world. But did you know that one is even more esoteric than the other? http://www.stat.columbia.edu/ gelman/research/published/diceRev2.pdf